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Abstract

To address the concurrent demands for precision and efficiency in wood surface defect detection, this study proposes a lightweight YOLO11n framework enhanced with attention mechanisms. We systematically integrated five distinct attention modules into the Backbone and Neck sections of the model, constructing a series of comparative "YOLO11n + Attention" models to investigate the impact of different mechanisms and embedding strategies. Furthermore, a novel three-dimensional weighted evaluation method incorporating accuracy, speed, and resource consumption was proposed for comprehensive model assessment. Experimental results identified the SEAM attention mechanism as the top performer. The YOLO11n+C2PSA_SEAM model achieved the highest detection accuracy, with a mAP@0.5 of 90.4%, a 3.6 percentage-point improvement over the baseline (86.8%). Notably, the YOLO11n+SEAM model ranked first in the comprehensive evaluation, maintaining a high inference speed of 476 FPS and achieving the best computational efficiency. The study also demonstrated that embedding the attention module within the Backbone yields superior overall performance compared to the Neck. This work provides a high-precision, efficient, and practically evaluable lightweight solution for automated defect detection in the wood industry.

Keywords： Wood defect detection, YOLO11n, Attention mechanism, Three-dimensional weighted comprehensive evaluation

1 Introduction

Wood is a key raw material for construction and furniture manufacturing, and its surface quality directly affects product performance and resource utilization. However, during growth, processing, and storage, it is prone to defects such as knots, cracks, and resin pockets(Wang & Yang, 2020). Traditional manual inspection is subjective and inefficient, with accuracy often below 70% (Chen and Zhang, 2021), failing to meet modern industry's high-precision and real-time requirements, thus reducing wood utilization (FAO, 2023). Although computer vision has been widely applied in defect detection (Liu and Kamal, 2021; Zhang et al., 2019), challenges remain in balancing speed and accuracy due to complex textures and diverse defect shapes (Mohsin et al., 2022; Kober and Alvarez, 2020). Deep learning, with end-to-end feature learning (Tu et al., 2023), has improved detection accuracy and ensured real-time performance.

With AI advancement, machine learning and deep learning have become mainstream in wood defect detection. In machine learning, Mohsin et al. (2019) applied multi-channel Gabor filters and

k-means clustering for segmentation. In CNN-based approaches, Hu et al. (2021) designed a tree-type CNN to classify live and dead knots with over 90% accuracy, while Cheng et al. (2022) improved ResNet, achieving an mAP of 89.7% in wormhole detection. Lin et al. (2017) proposed FPN to mitigate small-scale defect feature loss.

In object detection, two-stage methods are represented by Faster R-CNN (Ren et al., 2015), which introduced the RPN and laid the foundation for industrial applications. One-stage methods are represented by YOLO: Redmon et al. (2016) proposed the first real-time single encoder-decoder model. In wood defect detection, the YOLO series has evolved: Cao et al. (2024) proposed YOLOv5-LW with ECA and MBiFPN, improving feature processing but reducing speed; Yan et al. (2025) proposed YOLOv8-CBW, improving mAP50 by 3.7% and mAP50-95 by 3.9%, though its robustness under extreme conditions remains limited.

Despite these advances, three issues persist: difficulty balancing accuracy, efficiency, and lightweight design; insufficient optimization for specific defects (e.g., wave, edge_knot); and a lack of systematic comparison of attention mechanisms. Attention mechanisms are a promising solution, but quantitative evaluation remains scarce.

This paper selects five representative attention mechanisms—Efficient Multi-Scale Attention (EMAttention) (Ouyang et al., 2023), Vision Transformer with Bi-Level Routing Attention (BiFormer) (Zhu et al., 2023), Triplet Attention (Misra et al., 2021), Multi-Scale Dilated Attention (MSDA) (Jiao et al., 2023), and Separated and Enhancement Attention Module (SEAM) (Yu et al., 2022)—and embeds them into the YOLO11 framework to systematically evaluate their optimization effects in wood defect detection.

The main contributions of this work are as follows:

(1) In response to the requirements of wood defect detection, we systematically compare the embedding effects of five attention mechanisms in the YOLO11 Backbone and Neck, providing empirical evidence for embedding strategies;

(2) We propose a three-dimensional weighted evaluation method for industrial applications, which integrates seven indicators across three dimensions: accuracy (mAP50, mAP50-95, Precision, Recall), speed (FPS), and resource consumption (Params, FLOPs). By applying differentiated weighting configurations, this method balances accuracy, real-time performance, and deployment cost, offering a quantitative basis for model selection;

(3) Experimental results demonstrate that SEAM achieves the best overall performance in YOLO11, significantly improving accuracy while maintaining low computational overhead, making it particularly suitable for edge-device deployment;

(4) Comparative analysis of embedding positions shows that attention modules embedded in the Backbone (especially when enhanced with C2PSA) generally outperform those embedded in the Neck, providing important guidance for subsequent structural optimization.

2 Materials and methods

This section delineates the experimental setup devised to evaluate the performance of attention-enhanced YOLO11 models for wood defect detection. It begins with a description of the dataset and data augmentation techniques. Then, it details the baseline YOLO11 architecture and the five integrated attention mechanisms. Finally, it explains the two primary strategies for embedding these mechanisms and outlines the implementation details for model training and evaluation.

2.1 Data

This study utilized the Oulu-DET wood defect dataset compiled by Central South University. Oulu-DET originated from the University of Oulu and was cleaned and re-labeled by Central South University. It consists of 839 images of spruce wood and was captured using a controlled halogen lamp with a 3200K color temperature. Each image is in RGB color, with 8 bits for each channel, and has a size of 512×488 pixels. The error in illumination variation is kept within 1%. The original data is in PASCAL VOC format. To adapt to the training of YOLO series models, all annotations in this study were uniformly converted to YOLO format. The dataset includes six types of defects: dry knots, sound knots, edge knots, small knots, cracks, and waves. Some sample illustrations are shown in Fig.1.

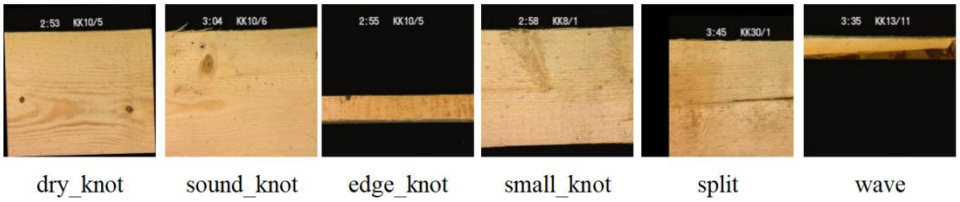


Fig.1 Partial Wood Defect Sample Images

To enhance the generalization ability of the model, three data augmentation strategies were implemented: geometric transformation (rotation, flipping, affine distortion), photometric distortion (brightness, contrast adjustment, and Gaussian noise), and local blurring. The preprocessing results are shown in Fig. 2. After augmentation, the number of images increased to 3785, which were divided into training set (3028 images), validation set (377 images), and test set (380 images) in a ratio of 8:1:1.

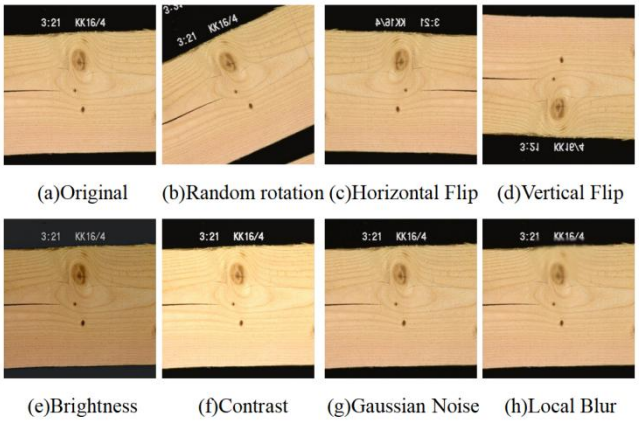


Fig.2 Data augmentation examples: (a) Original, (b) Random rotation, (c) Horizontal flip, (d) Vertical flip, (e) Brightness adjustment, (f) Contrast adjustment, (g) Gaussian noise, (h) Local blur.

2.2 Methods

2.2.1 YOLO11

This study adopts YOLO11n as the baseline model, which is a new-generation multi-task visual detection architecture introduced by Ultralytics in 2024. While maintaining high efficiency and real-time performance, it optimizes the structure, training strategy, and multi-task coordination mechanism through modular design, making it particularly suitable for resource-constrained scenarios. The model follows the classic three-stage Backbone-Neck-Head architecture: the Backbone enhances feature extraction and semantic focus capabilities via C3k2 and C2PSA modules; the Neck introduces a learnable gating structure in the feature pyramid network to optimize multi-scale feature fusion; and the

Head adopts a decoupled design that balances classification efficiency and regression accuracy, achieving an effective trade-off in detection performance.

2.2.2 Introduce different attention mechanisms

In deep learning-based wood defect detection, enhancing the model's ability to perceive fine-grained and multi-scale defects remains a critical challenge. Attention mechanisms, by emphasizing key features and suppressing redundant information, provide an effective means of performance improvement. Based on preliminary experiments, this study selects five structurally novel and high-performing attention models, which are systematically compared after being embedded into YOLO11:

(1) EMAttention: Employs channel-batch reshaping and group convolution to reduce computational cost while preserving multi-scale feature modeling capability, making it suitable for tasks requiring both long-range dependency and local detail representation.

(2) BiFormer: Utilizes a bi-level routing mechanism to dynamically allocate computational resources, enabling efficient modeling of both global context and local details at low overhead, thus improving the perception of complex defect structures.

(3) Triplet Attention: Captures cross-dimensional interactions through a three-branch structure, enhancing stable recognition of fine defects such as knots and cracks without introducing additional parameters.

(4) MSDA: Applies multi-rate dilated self-attention with parallel feature aggregation, effectively addressing variations in defect scales and improving adaptability to defects of different sizes.

(5) SEAM: integrates depthwise separable convolution (with residual connections) for preprocessing and 1×1 convolution for channel fusion. It employs a fully connected layer to learn feature weights, which are exponentially scaled to amplify key responses. This lightweight design enhances subtle and multi-scale defect features while suppressing noise, making it ideal for low-contrast defect detection in YOLOv11 without compromising real-time performance.

2.2.3 Model Structure with Embedded Attention Mechanism

To fully leverage the attention mechanism to enhance the performance of the timber defect detection model, this study embedded attention modules into both the Backbone and Neck sections, systematically improving the model's capability in multi-scale feature extraction and fusion.

Within the Backbone structure, based on the YOLOv11n benchmark model, a secondary innovation strategy was adopted to improve the C2PSA module, which served as the core component for embedding different attention mechanisms. Taking SEAM as an example, it was integrated into the C2PSA module to construct the C2PSA_SEAM module. Its structural diagram is shown in Fig. 3.

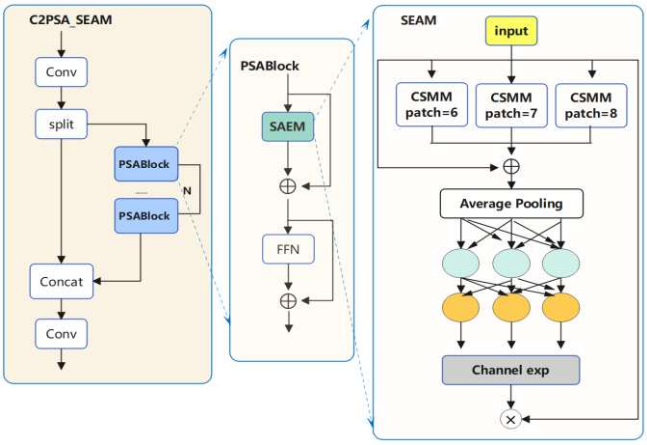
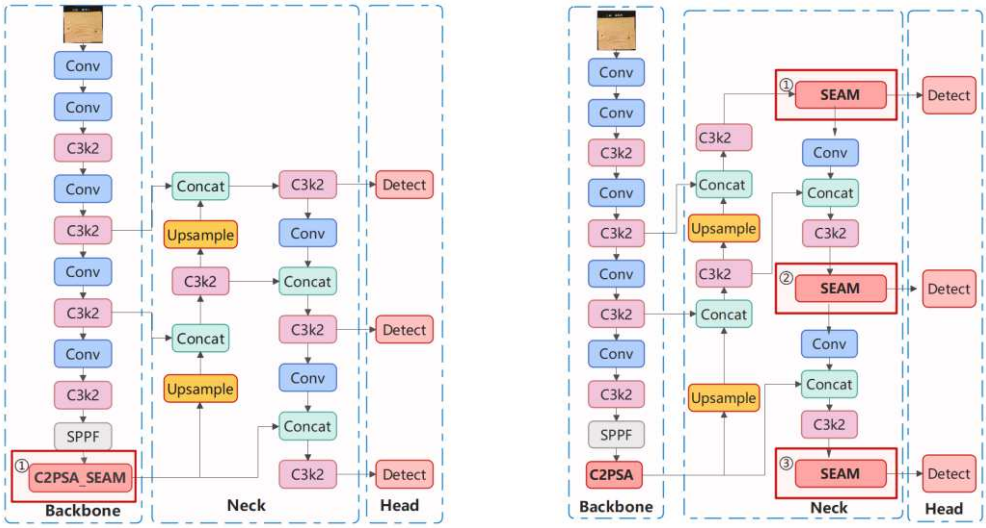


Fig. 3 Schematic of the C2PSA_SEAM, PSABlock, and SEAM modules (The structure of the SEAM module is adapted from Yu et al., 2022).

C2PSA_SEAM is built upon the C2PSA structure: the input feature map is first processed by a convolution and then split into two branches—one maintains an identity mapping, while the other undergoes deep enhancement through N PSABlocks incorporating the SEAM attention mechanism. The core of PSABlock is the SEAM module (Yu et al., 2022), which processes multi-scale image patches via a Cross-Scale Matching Module (CSMM), extracts spatial features using depthwise separable convolution, establishes channel dependencies, and adaptively enhances key regions through channel attention. The two branches are finally concatenated, followed by convolution for integration and dimensionality reduction, outputting optimized features. This structure is tailored to enhance the model's perception of subtle textures and irregular shapes typical in wood defects. The resulting detection model, named YOLO11n+C2PSA_SEAM, is illustrated in Fig.4 (a), with the new module indicated by a red box.



(a)YOLO11n + C2PSA_SEAM (b)YOLO11n+SEAM

Fig.4 Network Architecture Comparison Between YOLO11n + C2PSA_SEAM and YOLO11n + SEAM

In the Neck part, the SEAM module is embedded into different detection layers to strengthen multi-scale feature generalization: P3/8 (layer 17) focuses on small-scale defects such as fine cracks, P4/16 (layer 21) targets medium-scale defects, and P5/32 (layer 25) emphasizes large-scale semantic extraction. The constructed YOLO11n+SEAM model is depicted in Fig. 4 (b), with the newly added SEAM modules marked by red boxes 1–3. This hierarchical embedding strategy enhances the model's

robustness in detecting wood features of diverse scales and shapes, and the module positions can be flexibly adjusted according to the task.

3 Results and discussion

3.1 Implementation Details

The experiments in this study were conducted on the following software and hardware environment: The hardware platform utilized a workstation equipped with a 12 vCPU Intel Xeon Silver 4214R @ 2.40GHz CPU, an NVIDIA GeForce RTX 3080 Ti (12GB VRAM), and 45GB RAM. The software environment consisted of Ubuntu 22.04 operating system, Python 3.10.8 programming language, PyTorch 2.1.2 (CUDA 12.1) deep learning framework, with model development and training based on the Ultralytics 8.3.2 library.

The model training adopted the following key hyperparameters: momentum of 0.937, weight decay of 0.0005, and batch size automatically configured via the AutoBatch functionality (occupying approximately 60% of GPU VRAM). The initial learning rate (lr0) was set to 0.001, final learning rate (lrf) to 0.01, training epochs to 100, with the AdamW optimizer selected. All input images were uniformly resized to 640×640 pixels.

3.2 Evaluation Metrics

This paper selects evaluation metrics from three aspects: detection accuracy, computational efficiency, and generalization capability. In terms of accuracy, metrics include precision (P), recall (R), and mean average precision (mAP@0.5, mAP@0.5:0.95). Among these, mAP@0.5 serves as the primary metric, representing the average precision at IoU=0.5; mAP@0.5:0.95 evaluates robustness under multiple thresholds; precision and recall measure prediction accuracy and coverage rate, respectively. For efficiency, frames per second (FPS) is the core metric, calculated as the reciprocal of single-image processing time, reflecting real-time performance. Resource-related metrics include parameters (Params) and floating point operations (FLOPs), which quantify computational complexity and resource requirements to accommodate edge deployment. The calculation formulas for each metric are as follows:

$$p = \frac{TP}{TP + FP} \tag{1}$$

$$R = \frac{TP}{TP + FN} \tag{2}$$

$$AP = \int_0^1 p(r)dr \tag{3}$$

$$mAP = \frac{\sum AP_c}{N} \tag{4}$$

$$Params = \sum_{I=1}^{L} (K_I^2 \times C_{I,in} \times C_{I,out} + C_{I,out}) \tag{5}$$

$$FLOPs = \sum_{I=2}^{L} (K_I^2 \times C_{I,in} \times C_{I,in} \times H_I \times W_I + C_{I,out} \times H_I \times W_I) \tag{6}$$

Here, L represents the total number of convolutional layers, K_I denotes the kernel size of the I-th layer, $C_{I,in}$ and $C_{I,out}$ refer to the input and output channel numbers, respectively. The calculation of FLOPs reflects the computational overhead required for inferencing a single image,

Among them,H_I and $W_I$are the height and width of the feature map of the I-th layer.

3.3 Selection of the Baseline Model

This study adopts YOLO11n as the baseline model, which achieves an optimal balance between detection accuracy and computational efficiency, making it particularly suitable for industrial inspection scenarios with high real-time requirements. As shown in Table 1, a comparison with YOLO11m and YOLO11n reveals that although YOLO11m delivers higher detection accuracy (mAP50: 93.9%), YOLO11n achieves a significantly higher inference speed (416 FPS) with considerably fewer parameters (2.58M) and lower computational cost (6.3G FLOPs). This provides ample computational headroom for subsequent introductions of additional operations, such as attention modules.

Table 1 Experimental Results of YOLO11n and YOLO11m Models

Method	mAP50(%)	Params (M)	FLOPs (G)	FPS
YOLO11n	86.8	2.58	6.3	416
YOLO11m	93.9	20.03	67.7	217

3.4 A Study on the Impact of Integrating Different Attention Mechanisms

In wood defect detection, addressing challenges such as diverse defect morphologies, multi-scale variations, and complex backgrounds makes the introduction of attention mechanisms particularly valuable for enhancing feature discriminability. This study integrates five attention mechanisms—EMAttention, BiFormer, TripletAttention, MSDA, and SEAM—into the YOLO11 framework to systematically evaluate their effects on detection performance. Each mechanism is tailored to the characteristics of wood imagery: EMA strengthens fine-grained perception through cross-scale fusion; BiFormer reduces computational redundancy via dynamic sparse attention; Triplet Attention captures cross-dimensional interactions with a lightweight structure; MSDA adapts to multi-scale defect recognition; and SEAM enhances semantic saliency for low-contrast defects.

To systematically investigate the impact of integration strategies, this paper adopts two approaches: first, integrating attention modules into the Neck section to enhance multi-scale feature fusion capability; second, embedding the modules into the C2PSA module of the Backbone to improve fundamental feature extraction performance. A comparative analysis of experimental results is presented in Table 2.

Table 2 Experimental Results for YOLO11 Variants with Five Attention Mechanisms across Two Embedding Configurations

Architecture Variant	Method	Evaluation Metrics						
		Accuracy Metric				Resource Metric		Speed Metric
		mAP50 (%)	mAP50-95(%)	Precision (%)	Recall(%)	Params (M)	FLOPs (G)	FPS
Baseline	YOLO11n	86.8	53.7	82.8	81.2	2.58	6.3	417
Neck Integration	+Biformer	82.5	48.1	80.3	76.8	2.94	7.3	476
	+EMAttention	88.3	54.1	84.69	82.5	2.59	6.5	400
	+MSDA	87.8	54.5	83.2	81.8	2.94	7.1	303
	+SEAM	87.8	53.8	82.9	81.2	2.69	6.6	476
	+TripletAttention	89.2	55.1	83.8	84.2	2.59	6.6	385

Backbone Integration	+C2PSA_Biformer	88.9	54.3	84.9	84.2	2.6	8.9	384
	+C2PSA_EMAttention	88.6	53.9	84.2	83.5	2.54	6.4	416
	+C2PSA_MSDA	89.6	56.5	87.1	82.7	2.6	6.3	370
	+C2PSA_SEAM	90.4	55.3	86.5	85.2	2.56	6.3	435
	+C2PSA_TripleAttention	89.3	54.4	84.3	82.7	2.53	6.3	435

Comparative analysis of Table 2 reveals a clear advantage of backbone integration over neck integration. Models with attention mechanisms embedded in the backbone consistently achieve higher accuracy while maintaining competitive efficiency.

3.5 Three-dimensional weighted comprehensive evaluation

3.5.1 Establishment of Evaluation Dimensions and Weight Allocation

In industrial applications of wood defect detection (e.g., plywood knots, solid wood cracks, and insect damage), the core principles are "quality first, efficiency considered, and cost controllable." To achieve accurate identification, reduce material waste and rework rates, while matching production line rhythms and controlling deployment costs, this study proposes a multi-objective weighted scoring method tailored for industrial scenarios. It comprehensively evaluates model performance from three dimensions: accuracy (50%), speed (30%), and resource consumption (20%).

The weight assignments of this method are determined based on field feedback from production managers at multiple wood processing plants, combined with actual constraints of typical production lines. Accuracy, as the quality baseline, directly determines system usability and product qualification rates, thus assigned the highest weight (50%). Speed is a hard constraint to ensure real-time throughput and avoid production line bottlenecks, weighted at 30%. Resource consumption affects deployment costs and economic feasibility, particularly relevant for small and medium-sized enterprises, hence assigned a weight of 20%.

Within the accuracy dimension, weights are further disaggregated based on risk preferences in defect detection: mAP50 (20%) serves as the primary criterion for customer acceptance; Recall (15%) prioritizes reducing missed detections; mAP50-95 (10%) comprehensively evaluates model robustness; Precision (5%) suppresses false alarms. The speed dimension uses FPS (30%) as the key metric to ensure alignment with production line cycles. The resource dimension covers FLOPs (12%) and Params (8%), jointly measuring computational and storage efficiency to promote lightweight design and edge deployment. The weights of each metric are summarized in Table 3.

Table 3 Weight Allocation of Metrics

Dimension	Metrics	Weight	Description
Accuracy 50 %	mAP50	20 %	AP at IoU=0.5, key for customer acceptance
	Recall	15 %	Recall, critical for avoiding missed detections
	mAP50-95	10 %	AP across IoU 0.5-0.95, measures localization detail
	Precision	5%	Precision, reduces false positives
Efficiency 30 %	FPS	30 %	Frames per second, measures real-time inference throughput
Resources 20 %	FLOPs	12 %	FLOPs, computational cost (lower is better)
	Params	8 %	Parameters, model size factor (lower is better)

3.5.2 Metric Normalization and Comprehensive Scoring Method

Due to the significant differences in the dimensions of each metric, we normalize them to a uniform

scale. Positive indicators (e.g., mAP50, Recall, FPS) are normalized by dividing the actual value by an ideal value (or the maximum observed value for FPS), while negative indicators (e.g., Params, FLOPs) are normalized by dividing the minimum observed value by the actual value. The ideal values and thresholds are set based on industrial standards (e.g., mAP50≥85%, FPS≥60) to reflect practical requirements.

Based on the normalized indicators, the comprehensive score of the model is calculated using the weighted sum method. The formula is as follows:

$$Y = 0.2 * \text{norm}(\text{mAP50}) + 0.15 * \text{norm}(\text{Recall}) + 0.1 * \text{norm}(\text{mAP50} - 95) + 0.05 * \text{norm}(\text{Precision}) + 0.3 * \text{norm}(\text{FPS}) + 0.12 * \text{norm}(\text{FLOPs}) + 0.08 * \text{norm}(\text{Params}) \quad (7)$$

In the formula, norm() denotes the normalized metric value, and the coefficients correspond to their respective weights within the hundred-point scoring system, ensuring the scoring logic is transparent and traceable. This method achieves a sound balance among accuracy, efficiency, and resource consumption, making it suitable for model selection in industrial scenarios. The overall scores for each model, calculated using Equation (7), are presented in the "Score" column of Table 4.

Table 4 Comprehensive Scores of Each Variant Model

Architecture Variant	Model Name	Score	Rank	Score (Weight +5%)	Rank (Weight +5%)	Score (Weight -5%)	Rank (Weight -5%)
Baseline	YOLO11n	85.29	5	84.57	5	86.01	5
Neck Integration	+Biformer	84.05	7	83.01	9	85.08	6
	+EMAttention	84.36	6	83.86	6	84.86	7
	+MSDA	76.20	12	76.47	12	75.93	12
	+SEAM	88.19	1	87.22	2	89.15	1
	+TripletAttention	83.71	8	83.37	7	84.05	8
Backbone Integration	+C2PSA_Biformer	80.67	10	80.62	10	80.71	10
	+C2PSA_EMAttention	85.71	4	85.1	4	86.31	4
	+C2PSA_MSDA	83.49	9	83.21	8	83.77	9
	+C2PSA_SEAM	87.94	2	87.28	1	88.59	2
	+C2PSA_TripleAttention	87.22	3	86.48	3	87.96	3
Average Score	Average (Neck)	83.30	—	82.79	—	83.81	—
	Average (Backbone)	85.01	—	84.54	—	85.47	—

The comprehensive evaluation results indicate that YOLO11n+SEAM ranks first with a score of 88.19, achieving the best balance among accuracy, speed, and resource consumption. This model's accuracy (mAP50: 87.8%) exceeds the industrial threshold, and its inference speed of 476 FPS fully meets the real-time requirements of high-speed production lines. Moreover, its parameter count (2.69M) and computational load (6.6G) are comparable to the baseline, demonstrating good adaptability for edge deployment. YOLO11n+C2PSA_SEAM closely follows with a score of 87.94, standing out as the most accurate model (mAP50: 90.4%). Its precision and recall rates are 86.5% and 85.2%, respectively, making it particularly suitable for scenarios with stringent detection rate requirements. Its slightly lower overall score is mainly due to a relatively lower FPS (435), reflecting the inherent trade-off between accuracy and speed.

From an application perspective, YOLO11n+C2PSA_SEAM demonstrates stronger reliability in

identifying low-contrast defects, while YOLO11n+SEAM is more suitable for high-throughput real-time detection scenarios. Both models maintain well-controlled computational complexity, making them valuable for practical deployment. In real-world applications, the choice of model should be based on specific requirements: for precision-first scenarios, the backbone-integrated approach (YOLO11n+C2PSA_SEAM) is recommended, whereas for speed-first scenarios, the neck-integrated approach (YOLO11n+SEAM) is preferable.

Overall, according to the three-dimensional weighted comprehensive evaluation results, the model group with attention mechanisms integrated into the backbone (average score: 85.01) significantly outperforms the neck-integrated strategy (average score: 83.30) in overall performance. This indicates that introducing attention mechanisms during the feature extraction stage can more comprehensively enhance the model's defect recognition capability and overall efficacy.

3.5.3 Weight Sensitivity Analysis

To validate the robustness of the three-dimensional weighted scoring method, a weight sensitivity analysis was conducted by increasing and decreasing the accuracy weight by 5% respectively (with corresponding adjustments made to other weights), in order to observe the sensitivity of model rankings to weight variations. The recalculated scores based on the new weights (results shown in the last four columns of Table 4) indicate that a ±5% fluctuation in weights did not lead to significant changes in model rankings. YOLO11n+SEAM and YOLO11n+C2PSA_SEAM consistently maintained the top two positions across all three weighting scenarios, and the average score of models with Backbone embeddings remained higher than those with Neck embeddings, further confirming the rationality of the initial weight settings and the reliability of the conclusions. The performance improvement brought by the SEAM module remained consistent under different weighting conditions, demonstrating good robustness.

3.5.4 Discussion of the Optimal Model

To analyze the feature focusing efficiency of different attention mechanisms, Fig.5 compares the heat maps of the baseline model and three improved models. Relatively speaking, the baseline model YOLO11n is slightly inferior in defect localization ability; after introducing the SEAM module, YOLO11n+SEAM can more accurately focus on defect areas such as cracks and knots, significantly suppressing background interference. The YOLO11n+C2PSA_SEAM, which further integrates the C2PSA structure, performs the best. It can not only highlight the main defects but also perceive subtle features and contextual information more continuously and consistently. The results show that the SEAM mechanism effectively improves the defect localization ability, while C2PSA_SEAM further enhances the perception performance of complex defects and details, jointly improving the detection accuracy and robustness.

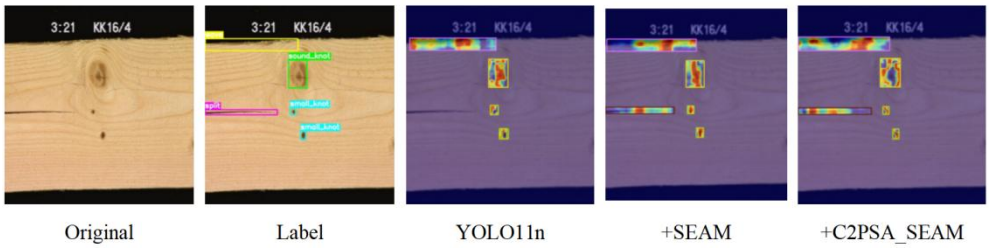


Fig. 5 Feature attention visualization: (a) Original input, (b) Ground truth labels, (c) Baseline YOLO11n heatmap, (d) YOLO11n+SEAM heatmap, (e) YOLO11n+C2PSA_SEAM heatmap.

As the most accurate model in this study, YOLO11n+C2PSA_SEAM demonstrates comprehensively superior performance on the validation set: it achieves an mAP@0.5 of 90.4%, with precision and recall rates of 86.5% and 85.2%, respectively—the highest among all compared models. The convergence behavior of different attention mechanism variants (Fig. 6) further confirms the superiority of this model: all models gradually stabilize after approximately 60 epochs, with the proposed variant demonstrating the highest and most stable mAP50 value (close to 0.91) in the final training stage. This indicates that its attention mechanism effectively enhances key feature extraction and noise suppression capabilities, providing an important basis for model selection in complex surface defect detection tasks.

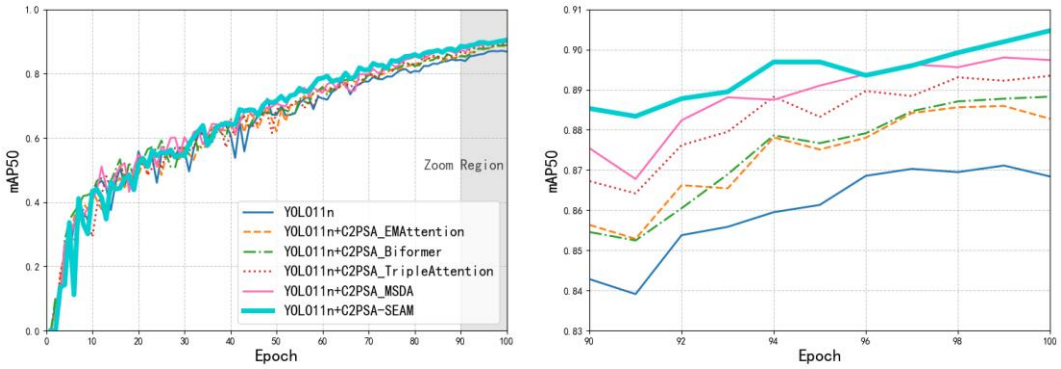


Fig. 6 The mAP50 change curves of different YOLO model variants: (left) full training progression; (right) enlarged view of the final 10 epochs (90-100).

To evaluate the performance improvement of the proposed C2PSA_SEAM attention module in wood defect detection, we conducted a quantitative comparison of the mAP50 performance between the improved model (YOLO11n+C2PSA_SEAM) and the baseline model (YOLO11n) across various defect categories. The specific results are presented in Table 5.

Table 5 Comparison of mAP50 Performance Between the Two Models Across Various Wood Defect Categories

Defect Class	YOLO11n (Baseline)	YOLO11n+C2PSA_SEAM	Absolute Improvement	Relative Improvement
dry_knot	0.932	0.948	+0.016	+1.7%
sound_knot	0.968	0.984	+0.016	+1.7%
edge_knot	0.916	0.966	+0.050	+5.5%
small_knot	0.833	0.860	+0.027	+3.2%
split	0.763	0.787	+0.024	+3.1%
wave	0.798	0.877	+0.079	+9.9%
all	**0.868**	**0.904**	**+0.036**	**+4.1%**

The comparative results demonstrate that the YOLO11n+C2PSA_SEAM model surpasses the baseline across all defect categories, achieving an overall improvement of 4.1% in mAP50. Notably, significant gains are observed for challenging slender or small-scale defects such as "wave" and "small_knot," with increases of 9.9% and 3.2%, respectively, highlighting its enhanced capability for subtle features. Moreover, the model exhibits balanced performance improvements across categories without significant scale bias, reflecting strong generalization ability. This progress is maintained even for categories like "sound_knot," where the baseline performance is already high. Quantitative validation further confirmed the model's excellent balance between precision and recall, achieving an

optimal F1-score of 0.85. These results collectively underscore the C2PSA_SEAM module's effectiveness in detecting diverse defects and its solid practical value.

To intuitively illustrate the predictive capability of the YOLO11n+C2PSA_SEAM model, Fig.7 compares the ground-truth labels (val_batch2_labels, left) with the corresponding predictions (val_batch2_pred, right) from a validation batch. The predictions show high consistency with the labels. The model not only accurately identifies most defect targets with minimal missed detections but also recognizes subtle features such as texture anomalies and discolorations, demonstrating robust feature learning and generalization in defect localization.

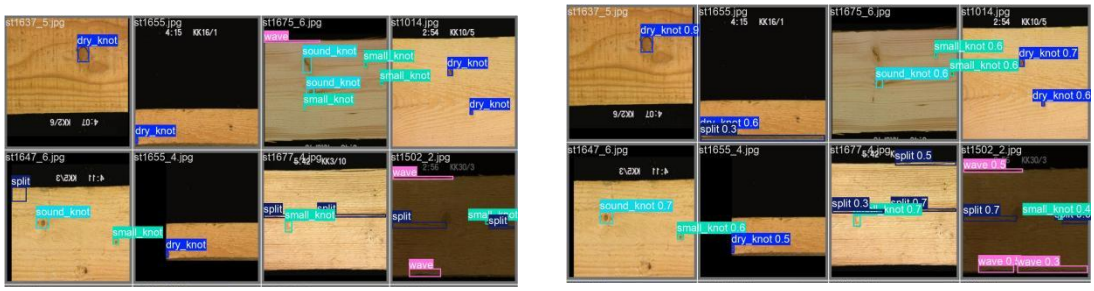


Fig. 7 Ground Truth (val_batch2_labels) vs. Model Predictions (val_batch2_pred)

4 Conclusion

This study systematically developed a series of YOLO11n models enhanced with various attention mechanisms for multi-scale wood surface defect detection and proposed a three-dimensional weighted comprehensive evaluation method integrating accuracy, speed, and computational resource consumption. The main conclusions are as follows:

The SEAM attention mechanism demonstrated superior performance, significantly improving the model's perception of low-contrast defects and its background suppression capability. Among the variants, the YOLO11n+C2PSA_SEAM model achieved the highest detection accuracy, reaching an mAP@0.5 of 90.4%, which represents a 3.6% improvement over the baseline model, with performance gains observed across all defect categories. In terms of overall comprehensive performance, the YOLO11n+SEAM model ranked first. It maintains high inference speed (476 FPS) and a low parameter count (2.69M) while exhibiting excellent detection capability for defects of various scales, achieving an optimal balance between accuracy and efficiency.

Experiments indicated that embedding the SEAM mechanism into the Backbone is more effective for enhancing feature extraction than integrating it into the Neck. The proposed three-dimensional weighted evaluation method offers good generality and transferability. Its weights can be flexibly adjusted according to specific application scenarios (e.g., safety requirements, throughput demands), providing a reliable and quantifiable decision-making framework for model selection in industrial visual inspection tasks.

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Data Availability: The Oulu-DET wood defect dataset used in this study is publicly available via Baidu Netdisk: https://pan.baidu.com/s/11Ft2fnZdsqmZWP6d6aRNnw (Access code: 9h3l). The code generated during the current study is available from the corresponding author on reasonable request.

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Compliance with Ethical Standards:

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Not applicable.

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